

Software Requirements Specification (SRS)

Project: Collaborative Study Material & Quiz Platform

Version: 1.1 (Revised for feasibility)

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Date: -

Status: Draft

Revision history

Version	Date	Author	Change summary	Approval
1.0	02-09-2026	Team 10	Initial SRS for Collaborative Study Material & Quiz Platform	Pending
1.1	03-09-2026	Team 10	Scaled down NFR-001 and NFR-003 to be verifiable without dedicated load-testing infrastructure	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	-	-	-

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Collaborative Study Material & Quiz Platform. The purpose of the system is to provide students with a centralized platform for sharing study materials, maintaining question banks, creating and attempting quizzes, and analysing academic performance. This document defines the functional requirements, non-functional requirements, system interfaces, security requirements and acceptance criteria for the proposed system.

1.2 Scope

The Collaborative Study Material & Quiz Platform is a web-based application that enables students to upload, access and share academic study materials. Users can create and attempt quizzes, contribute questions to a common question bank and view their quiz performance. Administrators can moderate inappropriate or invalid study materials, questions and quizzes. The system also provides simple subject-wise performance analysis based on quiz attempts and scores.

1.3 Audience

- Students
- Developers
- Testers
- Project Guide / Instructor
- System Administrators
- Project Evaluators

1.4 Definitions

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface

Term	Definition
CRUD	Create, Read, Update and Delete
Quiz	A collection of questions attempted by users
Question Bank	Repository containing academic questions
Study Material	PDF notes or other academic resources uploaded to the platform
Admin	Authorized user responsible for managing the platform
MERN	MongoDB, Express.js, React.js and Node.js

2. Overall description

2.1 Product perspective

The Collaborative Study Material & Quiz Platform is a web-based educational application intended to provide a common space for students to share learning resources and evaluate their knowledge through quizzes. The platform consists of a user-facing web application, backend services and a database. Users interact with the system through a browser while the backend handles authentication, study material management, quiz management, question-bank operations and performance analysis.

2.2 Major product functions (detailed)

- User registration and login
- Upload study materials
- Search and access study materials
- Download study materials
- Create questions
- Maintain a question bank
- Create quizzes
- Attempt quizzes
- Automatically evaluate objective questions
- Display quiz results
- View quiz attempt history
- View subject-wise average performance
- Moderate inappropriate study materials
- Moderate inappropriate questions and quizzes

- Search and filter quizzes and resources

2.3 User roles and characteristics (expanded)

- **Student:** Can register, log in, upload and access study materials, contribute questions to the question bank, create and attempt quizzes, and view performance analysis.
- **Administrator:** Can moderate study materials, questions and quizzes, and maintain the platform.

2.4 Operating environment

The proposed web application will use the MERN stack consisting of MongoDB, Express.js, React.js and Node.js. The system will run as a web application on modern desktop and mobile browsers, developed and demonstrated on a single local/dev deployment (no distributed infrastructure required).

2.5 Constraints

- Users must have internet connectivity to access the web application.
 - Users must register before accessing protected functionality.
 - Uploaded study materials must be PDF files and follow the configured file-size restriction.
 - The system shall use MongoDB as the primary database.
 - The web application shall be compatible with commonly used modern browsers.
 - The project must be completed within the academic project timeline using a small team and shared coursework schedule; NFRs are scoped to be verifiable through manual testing rather than dedicated load-testing infrastructure.
 - User passwords and sensitive information must be securely stored.
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3. External interface requirements

3.1 User interfaces

The system shall provide a responsive and user-friendly graphical web interface:

- Login page
- Registration page
- Student dashboard
- Study material page
- Upload material page
- Question bank page
- Quiz creation page
- Quiz attempt page

- Quiz results page
- Performance dashboard
- Administrator dashboard

The interface shall provide clear navigation menus, validation messages and appropriate success or error notifications.

3.2 Hardware interfaces

The system does not require specialized hardware. Users shall access the platform using devices such as desktop computers, laptops, tablets or smartphones with internet connectivity.

3.3 Software interfaces

- MongoDB database for storing users, materials, questions, quizzes and performance information.
- REST APIs for communication between frontend and backend.
- React.js frontend for displaying the user interface.
- Node.js/Express.js backend for implementing business logic and APIs.

3.4 Communications

Communication between the client and server shall use HTTP/HTTPS protocols. REST-based APIs shall be used for communication between the frontend and backend. HTTPS shall be used when the application is deployed to protect user credentials and other sensitive information during transmission.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow CSMQ-F-####.

4.1 Authentication

Description: Authenticate registered users and provide secure access to the platform.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQ-F-001	The system shall allow a new user to register using name, email	Functional	High	Student	AC-CSMQ-F-001: Valid and unique registration details create	Database required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	address and password.				a user account successfully. Test: TC-Auth-01	
CSMQ-F-002	The system shall allow registered users to log in using valid credentials and reject invalid credentials with an appropriate error message.	Functional	High	Student	AC-CSMQ-F-002: Valid credentials create an authenticated session; invalid credentials are rejected and an error message is displayed. Test: TC-Auth-02	Registered account required
CSMQ-F-004	The system shall allow an authenticated user to log out and terminate the active session.	Functional	High	Student	AC-CSMQ-F-004: After logout, protected pages cannot be accessed without logging in again. Test: TC-Auth-03	Active session required

4.2 Study Material Management

Description: Allow students to upload, search and access academic study materials.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQ-F-010	The system shall allow authenticated users to upload PDF study materials with a title, subject and description, subject to the configured maximum file size.	Functional	High	Student	AC-CSMQ-F-010: A valid PDF within the file-size limit is uploaded successfully; non-PDF or oversized files are rejected with an appropriate message. Test: TC-MAT-01	Authentication and storage required
CSMQ-F-012	The system shall allow users to search and filter study materials using keywords or subjects.	Functional	High	Student	AC-CSMQ-F-012: Matching study materials are displayed for the selected search or filter criteria. Test: TC-MAT-02	Material data required
CSMQ-F-013	The system shall allow users to view and download	Functional	High	Student	AC-CSMQ-F-013: Selected study material can	Material must exist

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	available study materials.				be opened or downloaded successfully. Test: TC-MAT-03	

4.3 Question Bank

Description: Maintain a collection of academic questions that can be searched and used while creating quizzes.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQ-F-020	The system shall allow authenticated users to add objective/MCQ questions with options, correct answer, subject, topic and difficulty level to the question bank.	Functional	High	Student	AC-CSMQ-F-020: A valid MCQ and its subject, topic and difficulty level are stored successfully in the question bank. Test: TC-QB-01	Authentication required
CSMQ-F-022	The system shall allow users to search and filter questions by subject, topic	Functional	High	Student	AC-CSMQ-F-022: Questions matching the selected	Question bank data required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	or difficulty level.				subject, topic or difficulty criteria are displayed. Test: TC-QB-02	

4.4 Quiz Management

Description: Allow users to create, search, attempt and evaluate quizzes.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQ-F-030	The system shall allow authenticated users to create a quiz by specifying a title and subject and selecting questions from the question bank.	Functional	High	Student	AC-CSMQ-F-030: A quiz with valid details and selected question-bank questions is created and stored successfully. Test: TC-QUIZ-01	Authentication and question bank required
CSMQ-F-032	The system shall allow users to search and view available quizzes.	Functional	High	Student	AC-CSMQ-F-032: Quizzes matching the search criteria are	Quiz data required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
					displayed. Test: TC-QUIZ-02	
CSMQ-F-033	The system shall allow registered users to attempt available quizzes.	Functional	High	Student	AC-CSMQ-F-033: Selected quiz opens and allows the user to answer its questions. Test: TC-QUIZ-03	Authentication and quiz required
CSMQ-F-034	The system shall automatically evaluate submitted objective/MCQ responses and calculate the user's score.	Functional	High	Student	AC-CSMQ-F-034: Calculated score matches the correct responses in the quiz. Test: TC-QUIZ-04	Correct answers required
CSMQ-F-035	The system shall display the quiz result after successful submission and evaluation.	Functional	High	Student	AC-CSMQ-F-035: The calculated result is displayed after quiz submission. Test: TC-QUIZ-05	Evaluation required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQ-F-036	The system shall maintain a history of quizzes attempted by each user.	Functional	Medium	Student	AC-CSMQ-F-036: Completed quiz attempts appear in the user's quiz history. Test: TC-QUIZ-06	Completed quiz attempt required

4.5 Performance Analysis & Administration

Description: Provide academic performance analysis and administrator functions for managing the platform.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQ-F-040	The system shall display the user's average score for each subject based on previous quiz attempts.	Functional	High	Student	AC-CSMQ-F-040: Subject-wise average scores are calculated from stored quiz results and displayed correctly. Test: TC-PERF-01	Quiz history required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQ-F-041	The system shall allow an administrator to remove inappropriate or invalid study materials.	Functional	High	Administrator	AC-CSMQ-F-041: Selected material is removed and is no longer accessible. Test: TC-ADMIN-01	Administrator authentication required
CSMQ-F-042	The system shall allow an administrator to remove inappropriate questions or quizzes.	Functional	High	Administrator	AC-CSMQ-F-042: Selected question or quiz is removed and is no longer accessible. Test: TC-ADMIN-02	Administrator authentication required

5. Non-functional requirements (detailed)

NFRs below are scoped to be verifiable manually by the team, without requiring dedicated load-testing tools or infrastructure. IDs CSMQ-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
CSMQ-NF-001	Standard application pages (dashboard, material list, quiz attempt) shall load and respond within approximately 3 seconds	Performance	High	Verified by manually timing page loads and key actions (login, search, quiz submit) during team testing; no

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
	under normal single-user use on a typical broadband/campus connection.			dedicated load-testing tool required. Test: TC-Perf-01
CSMQ-NF-002	The system shall preserve successfully saved user, study material, question, quiz and result data after an application restart.	Reliability	High	Saved records remain available and correct after restarting the backend/server and reconnecting to the database. Test: TC-Rel-01
CSMQ-NF-003	The system shall remain functional with at least 10 users active at the same time (e.g. multiple team members using separate browser sessions/incognito windows simultaneously), without crashing or returning errors.	Scalability	Medium	Manual test: team members access the app concurrently from separate sessions/devices and perform typical actions (browse, upload, attempt quiz) without functional failure. Test: TC-Scale-01
CSMQ-NF-004	The system shall support recent versions of Google Chrome, Microsoft Edge and Mozilla Firefox.	Compatibility	Medium	Core functions (register, login, upload, quiz attempt) are manually verified on all three browsers. Test: TC-Compat-01
CSMQ-NF-005	The user interface shall provide clear navigation and understandable success, validation and error messages.	Usability	Medium	A user can register, access study material and attempt a quiz without external assistance, verified through a walkthrough with a test user. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

1. **Confidentiality:** Protect user credentials and personal information from unauthorized access or disclosure.
2. **Integrity and Access Control:** Prevent unauthorized access, modification or deletion of user accounts, study materials, questions and quizzes.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
CSMQ-SR-001	The system shall securely hash user passwords (e.g. using bcrypt) and shall not store passwords in plain text.	Security	High	Database inspection confirms that passwords are stored only as password hashes. Test: TC-Sec-01
CSMQ-SR-002	The system shall require authentication (e.g. via JWT/session token) before allowing access to protected functionality.	Security	High	Unauthenticated users are denied access to protected functions/API routes. Test: TC-Sec-02
CSMQ-SR-003	The system shall enforce role-based access control for administrator-only functionality.	Security	High	Student accounts cannot access administrator routes or UI. Test: TC-Sec-03
CSMQ-SR-004	The system shall validate and sanitize user input before processing or storing it.	Security	High	Invalid or malicious input (e.g. script injection, malformed data) is rejected or sanitized without affecting stored data. Test: TC-Sec-04
CSMQ-SR-005	The deployed system shall use HTTPS for sensitive communication between the client and server.	Security	High	Sensitive client-server requests are transmitted using HTTPS in the deployed environment. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

- **Exit criteria for acceptance:** All high-priority functional requirements implemented and verified, no critical NFR or security failures, and RTM shows all test cases passed.
 - **Acceptance test suites:** Authentication, Study Material Management, Question Bank, Quiz Management, Performance Analysis, Administration, Performance (manual timing), Reliability, Scalability (manual multi-session), Security, Compatibility, and Usability tests.
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7. System models and diagrams

7.1 UML Use-Case Diagram

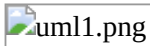


Diagram 1: Content & Question Bank

Actor: Student (+ Administrator connecting into “Remove Material” and “Remove Question/Quiz”)

Use cases:

- Register
- Login
- Logout
- Upload Study Material
- Search/View Study Materials
- Download Study Material
- Add Question to Bank
- Search/Filter Questions
- Remove Study Material (*Administrator*)
- Remove Question/Quiz (*Administrator*)

Relationships worth showing: `Upload Study Material` and `Add Question to Bank` both `<<include>>` a generic “Authenticate” use case (or just draw Login as a precondition); Administrator is a separate actor stacked above/below Student.

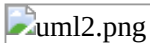


Diagram 2: Quiz & Performance

Actor: Student

Use cases:

- Create Quiz (selects from Question Bank — you can show this as a dependency arrow to Diagram 1 conceptually, or just note it in text)
- Search/View Quizzes
- Attempt Quiz
- Evaluate Quiz (*system auto-triggers this — often shown as `<<include>>` from Attempt Quiz*)
- View Quiz Result
- View Quiz History
- View Subject-wise Performance

Relationships worth showing: ; depends on quiz history data.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
CSMQ-F-001	User registration	4.1	AuthModule	TC-Auth-01	N	
CSMQ-F-002	User login	4.1	AuthModule	TC-Auth-02	N	
CSMQ-F-004	User logout	4.1	AuthModule	TC-Auth-03	N	
CSMQ-F-010	Upload/validate PDF material	4.2	MaterialModule	TC-MAT-01	N	
CSMQ-F-012	Search/filter material	4.2	MaterialModule	TC-MAT-02	N	
CSMQ-F-013	View/download material	4.2	MaterialModule	TC-MAT-03	N	
CSMQ-F-020	Add/classify MCQ question	4.3	QuestionBankModule	TC-QB-01	N	
CSMQ-F-022	Search/filter question	4.3	QuestionBankModule	TC-QB-02	N	
CSMQ-F-030	Create quiz with questions	4.4	QuizModule	TC-QUIZ-01	N	
CSMQ-F-032	Search/view quiz	4.4	QuizModule	TC-QUIZ-02	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
CSMQ-F-033	Attempt quiz	4.4	QuizModule	TC-QUIZ-03	N	
CSMQ-F-034	Evaluate MCQ quiz	4.4	EvaluationModule	TC-QUIZ-04	N	
CSMQ-F-035	Display result	4.4	EvaluationModule	TC-QUIZ-05	N	
CSMQ-F-036	Quiz attempt history	4.4	EvaluationModule	TC-QUIZ-06	N	
CSMQ-F-040	Subject-wise average score	4.5	PerformanceModule	TC-PERF-01	N	
CSMQ-F-041	Moderate materials	4.5	AdminModule	TC-ADMIN-01	N	
CSMQ-F-042	Moderate questions/quizzes	4.5	AdminModule	TC-ADMIN-02	N	
CSMQ-NF-001	Response time (manual check)	5	WebUI / Backend	TC-Perf-01	N	
CSMQ-NF-002	Data persistence	5	Backend / Database	TC-Rel-01	N	
CSMQ-NF-003	Concurrent users (10, manual)	5	Backend / Database	TC-Scale-01	N	
CSMQ-NF-004	Browser compatibility	5	WebUI	TC-Compat-01	N	
CSMQ-NF-005	Usability	5	WebUI	TC-UX-01	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
CSMQ-SR-001	Password hashing	5.1	AuthModule	TC-Sec-01	N	
CSMQ-SR-002	Protected access	5.1	AuthModule	TC-Sec-02	N	
CSMQ-SR-003	Role-based access	5.1	AuthModule / AdminModule	TC-Sec-03	N	
CSMQ-SR-004	Input validation	5.1	Backend	TC-Sec-04	N	
CSMQ-SR-005	HTTPS communication	5.1	WebUI / Backend	TC-Sec-05	N	